



Machine learning-driven prediction and interpretation of air quality index in industrial environment

R. Muralikrishnan¹ · Suneel Gollapalli² · Elayaraja Sellappan³ · J. Prasanya⁴ · V. Murali⁵ · A. Vijayakumar⁶

Received: 16 September 2025 / Accepted: 14 October 2025
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025

Abstract

Air pollution in South India's industrial clusters remains poorly studied, despite their growing importance as sources of pollutants in the region. This study presents an in-depth examination of air quality in Gummidipoondi, Tamil Nadu, encompassing statistical analysis, feature importance, and machine learning models. The results showed that PM_{2.5} and PM₁₀ are the main features that affect the Air Quality Index (AQI), accounting for more than 99% of its variations. Gaseous pollutants and weather conditions had only a lesser effect. XGBoost was the most accurate machine learning model ($R^2 = 0.9965$, RMSE=0.0573), compared to regression and neural models. The SHAP summary plot confirmed that PM_{2.5} is the main feature that had a significant impact on AQI. Temporal patterns indicated significant pollution impacts during winter and post-monsoon seasons, attributed to slow boundary layer conditions and daily industrial activities, while monsoon offered natural alleviation through dispersion. These results broaden air quality studies outside urban areas and illustrate the effectiveness of explainable machine learning for real-time surveillance and early warning systems. The study underscores the immediate necessity for focused particulate matter regulation to mitigate health hazards in industrialising corridors.

Keywords Air quality index · Machine learning · Particulate matter · Gaseous pollutants

Introduction

Air pollution is now one of the biggest problems for the environment and public health in the modern world (Manisalidis et al., 2020). The World Health Organisation reports that every year, millions of people die because of fine

particulate matter, ground-level ozone, and other pollutants. Air pollution in India has reached dangerous levels in many places. In cities and industrial areas, the amount of particulate matter is sometimes far higher than what is considered safe by national and international norms. The problem has worsened because of rapid industrialisation, urban growth,

✉ R. Muralikrishnan
murali660105@gmail.com

✉ Elayaraja Sellappan
elayaraja86@gmail.com

Suneel Gollapalli
suneel.g@gmrit.edu.in

J. Prasanya
prasanyajayabal@gmail.com

V. Murali
murali_v123@yahoo.co.in

A. Vijayakumar
vijayakumarkct@gmail.com

¹ Jei Mathaajee College of Engineering, Kanchipuram, Tamil Nadu 631552, India

² Department of Computer Science and Engineering, GMR Institute of Technology, Rajam, Srikakulam, Andhra Pradesh 532127, India

³ Department of Civil Engineering, PSG Institute of Technology and Applied Research, Coimbatore 641062, India

⁴ Dept of Civil Engineering, Karpaga Vinayaga College of Engineering and Technology, Chengalpattu, Tamil Nadu 603308, India

⁵ Department of Civil Engineering, Pandian Saraswathi Yadav Engineering College, Arasanoor, Sivagangai, Tamil Nadu 630 561, India

⁶ Department of Civil Engineering, V.S.B. Engineering College, Karur, Tamil Nadu 639 111, India

and rapid transportation (Manisalidis et al., 2020). This is especially true in peri-urban industrial corridors, where environmental monitoring and regulatory enforcement are generally less strict than in big cities.

The Air Quality Index (AQI) is a common term to measure and talk about the state of the air quality. Researchers, policymakers, and the general public can better understand air quality based on AQI, which combines measurements of several pollutants into one value with defined categories (Payus et al., 2022). The Central Pollution Control Board (CPCB) in India calculates AQI by considering pollutants such as $\text{PM}_{2.5}$, PM_{10} , SO_2 , NO_2 , O_3 , NH_3 , CO , and Pb . Fine particulate matter, especially $\text{PM}_{2.5}$, is often considered to be the most important cause of bad air quality. This is mostly because it can enter the respiratory system and cause many heart and lung problems.

Particulate matter is the most important factor in AQI calculation, and it also shows how different human-made sources affect the air quality. In India, some of the main sources of $\text{PM}_{2.5}$ and PM_{10} are industrial combustion processes, vehicle emissions, biomass burning, and construction activities (Joshi et al., 2025). Thermal power stations, steel processing units, and chemical businesses in peri-urban industrial clusters emit more particulate pollution (Zalakeviciute et al., 2020). Meteorological factors, namely Temperature, humidity, wind speed, and boundary layer dynamics, play a major role in the dispersion of the pollutants (Andújar-Maqueda et al., 2025). High winds and rain during monsoon seasons make it easier for pollutants to spread, while stagnant conditions in winter months keep contaminants close to the ground. Because of this complexity, it is hard to forecast air quality using typical statistical methods that presume linear connections (Mujtaba et al., 2025).

In recent years, the world has seen remarkable progress in technology and computer science. Among these developments, artificial intelligence (AI) has emerged as a leading field, drawing global attention. Unlike many other disciplines, AI focuses on creating machines and systems capable of demonstrating intelligence, learning independently, and making decisions (Kaveh, 2024). Machine learning is used to model the non-linear and high-dimensional relationship between contaminants and weather conditions. Machine learning determines the hidden patterns, adjusts to complicated interactions, and makes more accurate predictions than conventional regression or time-series methods (Tao et al., 2023). Ensemble-based models like Random Forest and Extreme Gradient Boosting (XGBoost) are great for estimating air quality because they can mix many decision trees and lower the problem of overfitting (Meena et al., 2024). Support Vector Regression (SVR) and artificial neural networks, such as multi-layer perceptrons, are becoming

more common in this field because they can find complicated, non-linear relationships between environmental data (Sananmuang et al., 2024). Importantly, combining machine learning with explainable AI frameworks like SHAP (SHapley Additive exPlanations) not only makes predictions more accurate but also makes them easier to understand, which is important for making decisions.

Several studies have shown that machine learning can be used to predict air quality on a worldwide scale. Research in China, South Korea, and other East Asian areas has shown that tree-based ensemble models can always do better than linear models at predicting $\text{PM}_{2.5}$ levels. For example, researchers in Beijing used XGBoost on large-scale monitoring datasets and got far better at making accurate predictions than with traditional approaches (Su et al., 2023). In India, similar trends have been observed, with Random Forest and gradient boosting techniques utilised in Delhi and Mumbai to analyse complicated pollutant-meteorology relationships (de Bont et al., 2024). Recent studies have emphasised the growing air pollution challenge across South Asian cities, linking particulate matter with severe health outcomes (Pant et al., 2024a, b). Risk-based assessments further highlight the importance of integrating predictive models with public health perspectives (Pant et al., 2023). In the industrial domain Pant et al., 2024a, b applied a Naïve Bayes classifier to predict $\text{PM}_{2.5}$ concentrations in an industrial cluster, underscoring the relevance of tailored machine learning approaches in non-urban contexts. Nonetheless, the majority of current research is predominantly focused on megacities, with relatively insufficient emphasis on peri-urban industrial clusters. These places are important to regional emission profiles because they have a high concentration of factories and energy production facilities, despite being less densely populated. They not only make the air quality worse in their own area, but they also add to pollution in nearby cities.

Most of the research done in India has been on big cities like Delhi, Bengaluru, Hyderabad, Mumbai and Chennai (K et al., 2024; Ravindiran et al., 2023, 2025). Unlike earlier studies that predominantly focused on urban megacities, this research provides new insights into peri-urban industrial environments such as Gummidipoondi, Tamil Nadu. The novelty lies in the integration of machine learning models with SHAP and LIME explanations to not only achieve highly accurate AQI predictions but also to provide transparent interpretability of pollutant contributions in an underexplored industrial context. This signifies a critical research gap, as these areas are substantial sources of particulate emissions yet do not receive equivalent levels of ongoing monitoring, public awareness, or research-informed policy measures. The current study rectifies these limitations by executing an extensive investigation of AQI

in Gummidipoondi, including statistical diagnostics, data transformations, temporal trend analysis, and machine learning-based prediction. The aim is to develop and evaluate explainable machine learning models for predicting the Air Quality Index (AQI) in an industrial environment. The objectives include (i) analysing temporal and seasonal patterns of air pollutants and meteorological variables in Gummidipoondi, (ii) applying and comparing different machine learning models for AQI prediction, (iii) integrating SHAP and LIME frameworks to enhance the interpretability of the results.

Materials and methods

Study area

The study was in Gummidipoondi, which is in Tamil Nadu, India, around 45 km north of Chennai. It is located in the Chennai Metropolitan Region (CMR) between 13.41°N latitude and 80.15°E longitude. There are many factories in the neighbourhood, including thermal power plants, steel

rolling mills, chemical facilities, and large-scale manufacturing units. These factories are a big source of pollutants in the area. Heavy traffic on National Highway 16 (part of the Chennai–Kolkata corridor) and freight traffic from adjacent ports make air pollution levels much worse. Gummidipoondi has a tropical wet and dry climate. Summers are hot (March to June, 30 to 40 °C), and there are two monsoon seasons: one in the southwest (June to September) and one in the northeast (October to December). Winters are mild (January to February, 20 to 28 °C). The average yearly rainfall is between 1,100 and 1,200 mm, with the north-east monsoon being the main source. Seasonal changes in wind patterns affect pollution levels. For example, during the monsoon months, stronger south-westerlies help spread pollutants, while weaker winds in the winter months might cause pollutants to build up. Gummidipoondi is a key place to investigate particulate-dominated air pollution and its effect on the Air Quality Index (AQI) because of the large amount of industrial activity, heavy traffic, and changing weather patterns. Figure 1 illustrates the study area map.

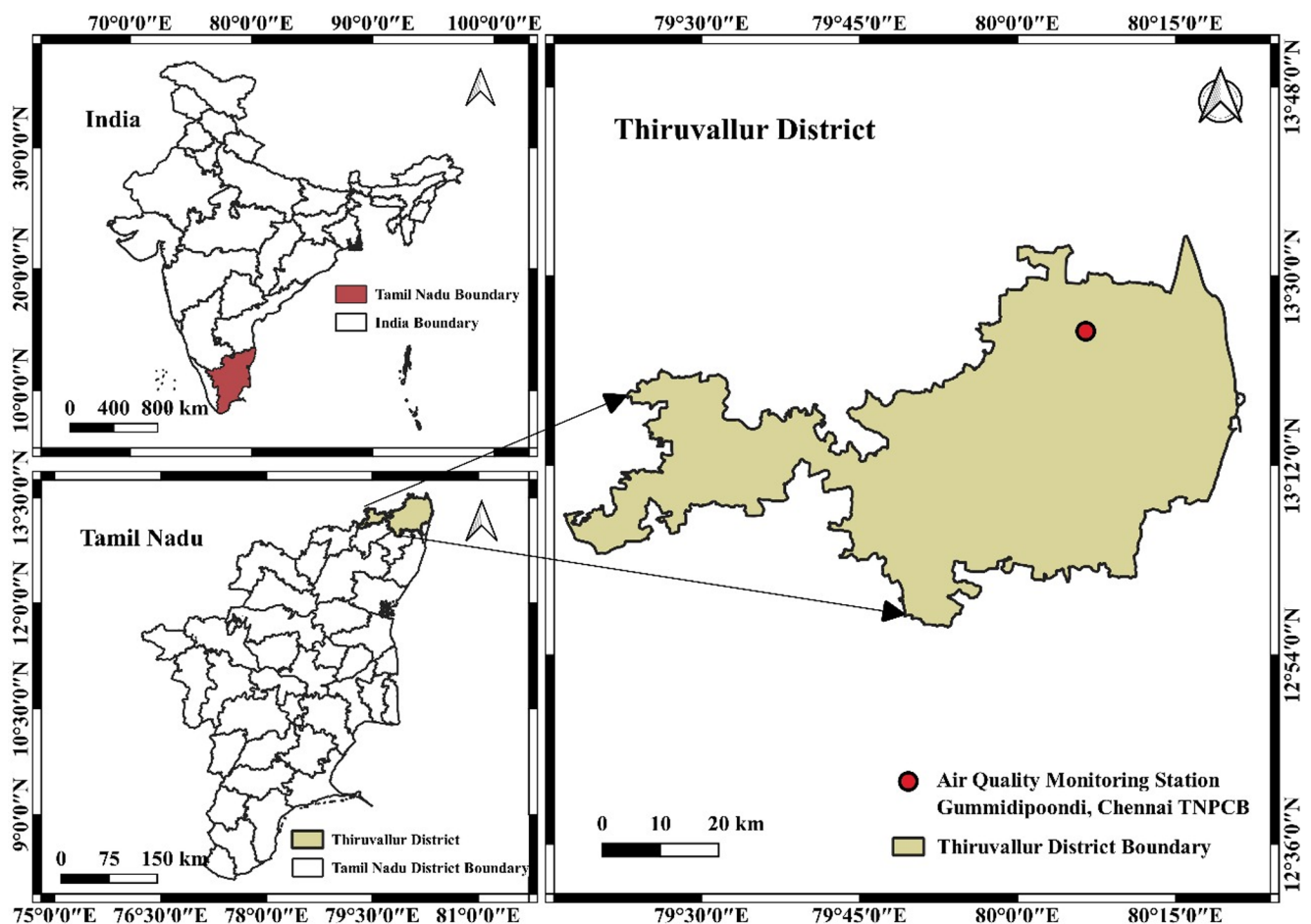


Fig. 1 Study Area Map

Dataset collection

The Central Pollution Control Board (CPCB) provided the dataset for this study through its Continuous Ambient Air Quality Monitoring Station (CAAQMS). There are 907 daily average records from 2023 to 2025 that include both air pollutants and weather conditions that are important for calculating AQI. The measured pollutants included PM_{2.5}, PM₁₀, NO, NO₂, NO_x, NH₃, and SO₂. The meteorological parameters were also recorded, were temperature, relative humidity, wind speed, sun radiation, and barometric pressure. To make the model stronger, pre-processing included dealing with missing values, and normalising skewed variables. Descriptive statistics showed that the average levels of PM_{2.5} (56.26 µg/m³) and PM₁₀ (105.05 µg/m³) are slightly higher than the safe standards, making particulate matter the most common type of pollution in the area. This dataset served as the basis for correlation analysis, evaluation of temporal variability, and machine learning-driven AQI prediction.

Data pre-processing

The CPCB dataset had daily average records of weather and air quality data. Before the analysis, the data were processed to make sure it was reliable and consistent. The missing values were removed from the datasets. The interquartile range (IQR) approach was used to find outliers, which are often caused by instrument faults or unexpected spikes. After pre-processing, we had a cleaned-up dataset of 907 valid records that was used for statistical and machine learning analysis.

The correlation analysis (Fig. 2) demonstrates that PM_{2.5} is the most significant factor affecting air quality, exhibiting a robust positive correlation with both PM₁₀ ($r=0.90$) and AQI ($r=0.97$), thus establishing that particulate matter is the primary contributor to pollution levels. Nitrogen oxides likewise show internal consistency, with NO substantially linked to total NO_x ($r=0.93$). However, their link to AQI is not as robust. Meteorological factors have substantial regulatory influences, with elevated air temperatures correlating with reduced PM_{2.5} concentrations ($r = -0.59$) and reduced AQI values ($r = -0.52$), indicating increased pollutant dispersion in warmer conditions. Wind speed also has a negative connection with NO ($r = -0.50$), which shows that stronger breezes disperse pollutants significantly. On the other hand, relative humidity has a negative correlation with temperature ($r = -0.73$) and a little positive correlation with PM_{2.5} ($r=0.26$). This suggests that humid conditions may help particles build up. Parameters like NH₃ and SO₂ show very weak correlations with AQI, which shows that they don't have a big effect on the overall air quality in this sample. The results show that PM_{2.5} is the most important

pollutant that changes the AQI, and that weather factors, including temperature, humidity, and wind, are also important in changing pollutant levels.

The descriptive statistics (Table 1) of the air quality dataset ($n=907$) show that the average PM_{2.5} concentration was 56.26 µg/m³, with values ranging from 8.3 to 190.55 µg/m³, which often exceeded the allowable limits. The average PM₁₀ concentration was 105.05 µg/m³, with peaks as high as 255.44 µg/m³, which shows that there is a lot of particulate pollution. NO and NO₂ had mean levels of 9.27 µg/m³ and 13.61 µg/m³, respectively, while NO_x had an average of 13.9 ppb. This suggests that vehicles or combustion are only contributing a small amount. The average level of ammonia (NH₃) was 13.43 µg/m³, which shows that there were significant emissions from farming or waste. The average level of SO₂ was 17.33 µg/m³, however, it sometimes went beyond 60 µg/m³. The weather data shows that the conditions are constant, with an average temperature of 31.57 °C, a high relative humidity of 73.48%, and a low wind speed of 1.71 m/s. These conditions are likely to cause pollutants to build up. The average solar radiation was 139.31 W/m², however, it varies based on season. The barometric pressure stayed around 764 mmHg.

Data transformation

Due to pollution events that happened from time to time, some variables, such as PM_{2.5}, PM₁₀, NO, NO₂, NO_x, and AQI, showed a lot of positive skewness and high kurtosis. The Yeo–Johnson transformation was used to make the distributions normal and the variance stable. Yeo–Johnson can handle both positive and zero values, which makes it a good choice for environmental datasets that sometimes have concentrations that are close to zero (Liu et al., 2024). This is different from the standard log or Box–Cox methods. After the transformation, the skewness was brought down to almost zero, and the kurtosis values were brought closer to Gaussian levels. This made the data more normal and ensured that the statistics were strong. This method made the dataset better for correlation analysis, regression modeling, and machine learning applications.

Before the transformation (Table 2), the skewness and kurtosis values showed that most air quality measures, including PM_{2.5}, PM₁₀, NO, NO₂, and AQI, were positively skewed with leptokurtic distributions. This means that they had extended right tails and were affected by very high levels of pollution. After using normalisation or transformation methods, the skewness values for these pollutants were brought down to almost zero, and the kurtosis values were brought closer to zero or slightly negative. This shows that the data is more symmetrical and normally distributed. The skewness of PM_{2.5} went from 1.20 to 0.001, and the kurtosis

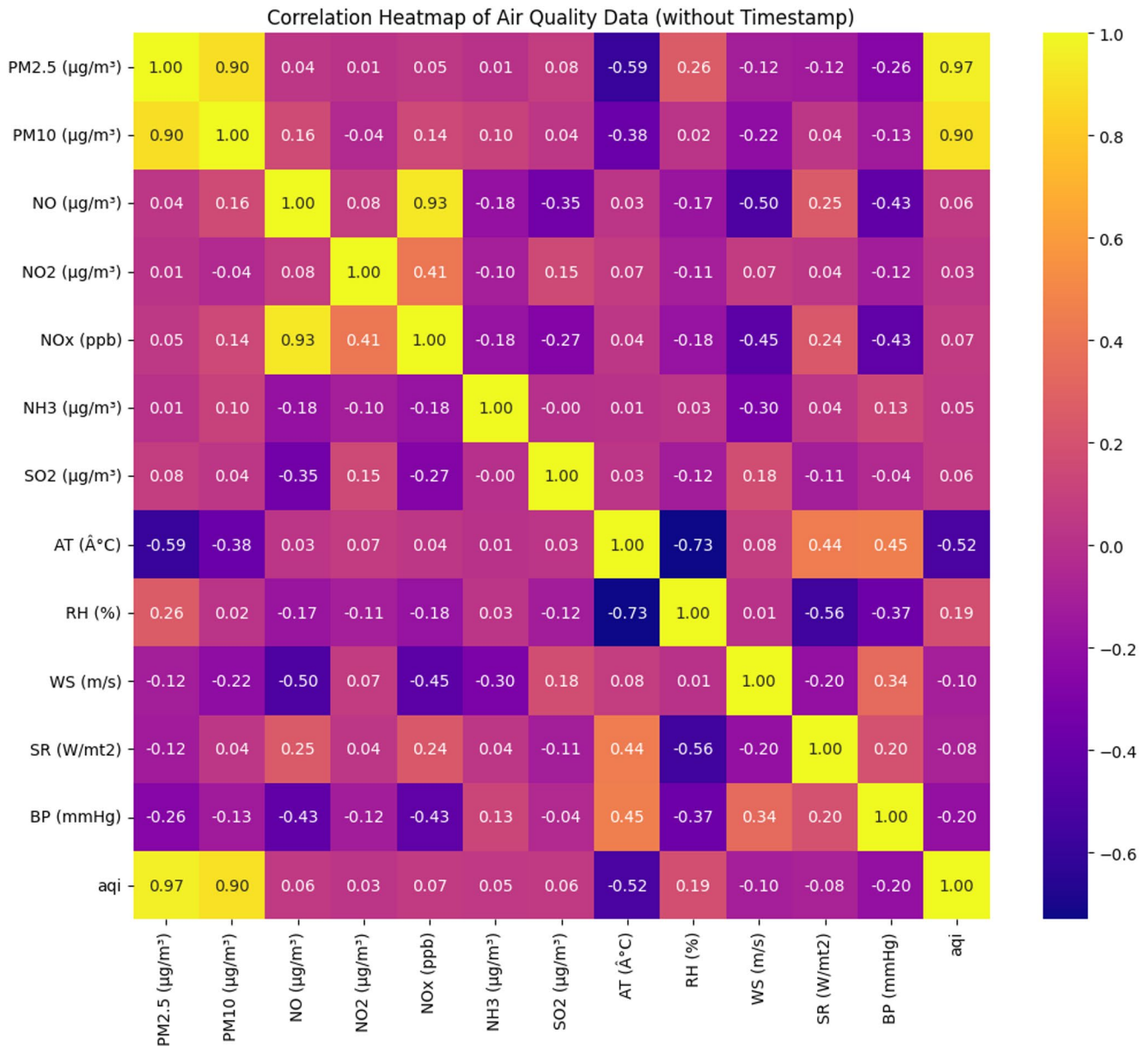


Fig. 2 Correlation matrix of pollutants and meteorological variables with AQI

Table 1 Descriptive statistics of air quality and meteorological parameters ($n=907$) at gummidipoondi

Input Features	Count	Mean	Std	Min	25%	50%	75%	Max
PM _{2.5} ($\mu\text{g}/\text{m}^3$)	907	56.26	28.33	8.3	35.88	49.89	70.57	190.55
PM ₁₀ ($\mu\text{g}/\text{m}^3$)	907	105.05	36.25	31.86	79.72	99	125.38	255.44
NO ($\mu\text{g}/\text{m}^3$)	907	9.27	11.74	0.1	2.42	4.12	9.48	55.09
NO ₂ ($\mu\text{g}/\text{m}^3$)	907	13.61	7.79	0.91	8.9	12.16	17.1	60.5
NOx (ppb)	907	13.9	10.97	0.44	6.25	10.49	16.33	56.16
NH ₃ ($\mu\text{g}/\text{m}^3$)	907	13.43	6.48	0.88	8.71	12.07	17.62	37.63
SO ₂ ($\mu\text{g}/\text{m}^3$)	907	17.33	11.11	0.79	8.77	15.46	23.48	60.92
AT ($^{\circ}\text{C}$)	907	31.57	1.7	27.28	30.2	31.63	32.84	38.65
RH (%)	907	73.48	6.93	49.86	69.47	73.09	77.27	98
WS (m/s)	907	1.71	0.99	0.3	0.7	1.82	2.42	6.59
SR (W/mt ²)	907	139.31	40.01	17	119.58	146.4	164.22	412.13
BP (mmHg)	907	763.96	3.71	743.19	761.85	763.92	765.72	774.21
AQI	907	121.11	64.11	31.86	80.37	100.82	136.77	354.27

Table 2 Skewness and kurtosis of pollutants and meteorological variables before and after Yeo–Johnson transformation

Input Features	Before Transformation		After Transformation	
	Skewness	Kurtosis	Skewness	Kurtosis
PM _{2.5} (µg/m ³)	1.20014	1.59958	-0.001	-0.2444
PM ₁₀ (µg/m ³)	0.87375	1.04772	-0.0004	-0.0194
NO (µg/m ³)	1.82568	2.19158	0.05451	-0.3219
NO ₂ (µg/m ³)	1.40217	3.30045	0.03381	0.60018
NO _x (ppb)	1.46946	1.56483	0.00071	0.03506
NH ₃ (µg/m ³)	1.02305	0.92612	0.01562	0.29817
SO ₂ (µg/m ³)	0.95808	0.75352	-0.036	-0.5165
AT (°C)	0.18903	-0.257	0.20371	-0.2362
RH (%)	0.29229	0.82559	0.285	0.82282
WS (m/s)	0.2795	-0.367	0.28676	-0.345
SR (W/mt ²)	-0.1046	3.38338	0.26071	4.73359
BP (mmHg)	-0.6789	4.76344	0.08085	2.06513
AQI	1.57606	1.92056	0.01422	-0.0359

Table 3 Hyperparameter tuning results for Ridge, random Forest, XGB, SVR, KNN, and MLP regressors

Model	Hyperparameter Tuning
Ridge	{‘alpha’: 10.0}
RandomForestRegressor	{‘max_depth’: 10, ‘min_samples_split’: 2, ‘n_estimators’: 200}
XGBRegressor	{‘learning_rate’: 0.1, ‘max_depth’: 7, ‘n_estimators’: 100}
SVR	{‘C’: 0.1, ‘epsilon’: 0.1, ‘kernel’: ‘linear’}
KNeighborsRegressor	{‘n_neighbors’: 5, ‘weights’: ‘distance’}
MLPRegressor	{‘activation’: ‘tanh’, ‘hidden_layer_sizes’: (50,), ‘solver’: ‘adam’}

from 1.599 to -0.244. The skewness of NO went from 1.825 to 0.054, and the kurtosis from 2.19 to -0.321. This shows that the distributional qualities improved a lot. The AQI also showed similar normalisation effects, with skewness going from 1.576 to 0.014 and kurtosis going from 1.92 to -0.035. On the other hand, climatic variables like air temperature, humidity, and wind speed didn’t alter much because their initial distributions were already closer to normal. Overall, the transformation technique worked well to reduce non-normality in pollutant data, making sure that the data was statistically sound for later analysis and modelling.

Hyperparameter tuning

The hyperparameter tuning (Table 3) procedure made each machine learning model work better by finding the greatest balance between accuracy and generalisation (Raiaan et al., 2024). For Ridge regression, the best parameter was set at $\alpha=10.0$, which mean that moderate regularisation was needed to keep the model stable and avoid overfitting. The ideal setup for RandomForestRegressor was 200 estimators,

a maximum depth of 10, and a minimum sample split of 2. This lets the model find non-linear relationships without making things too complicated. The tuning for XGBRegressor used 100 boosting iterations, a maximum depth of 7, and a learning rate of 0.1. This was done to find the right mix between model depth and learning speed to make sure the model was accurate and converged quickly. The Support Vector Regressor (SVR) worked best with a linear kernel, $C=0.1$, and $\epsilon=0.1$. This shows that a linear margin-based method with minimal regularisation is best for this dataset. For KNeighborsRegressor, the best setting was $k=5$ neighbours with distance-based weighting. This let the model give more weight to points that were closer together, which improved the accuracy of local predictions. Lastly, the MLPRegressor worked best with one hidden layer of 50 neurons, tanh activation, and the Adam solver. This shows how useful a shallow but non-linear network can be for capturing interactions between pollutants and AQI.

Machine learning models

Machine learning models, namely Ridge, RandomForestRegressor, XGBRegressor, Support Vector Machine (SVR), KNeighborsRegressor and MLPRegressor, were used for the prediction of AQI. Model Performance metrics were assessed using the parameters, namely Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2) (Kaveh et al., 2018; Kaveh & Khavaninzadeh, 2024; Ravindiran et al., 2025).

Results and discussion

Diurnal and seasonal variations

The daily changes in pollutants and weather at study area, show clear patterns (Fig. 3) throughout time that are affected by human activity and the weather. PM_{2.5} and PM₁₀ levels are highest in the early morning (0–8 h) and late at night (20–23 h), and they drop sharply in the middle of the day. This is because emissions are lower and dispersion is better when solar heating and mixing are stronger (Singh et al., 2020). NO, NO₂, and NO_x levels are highest in the morning (6–9 AM) and again in the evening, when there is a lot of traffic and industrial activity. They are lower in the afternoon because they are mixed with other gases (Kalbarczyk & Kalbarczyk, 2020). NH₃ stays rather steady, with little rises in the morning that are probably due to emissions in the area. SO₂ levels are highest in the early morning (4–8 AM) and then slowly drop throughout the day. This suggests that factories are burning at night and in the early morning.

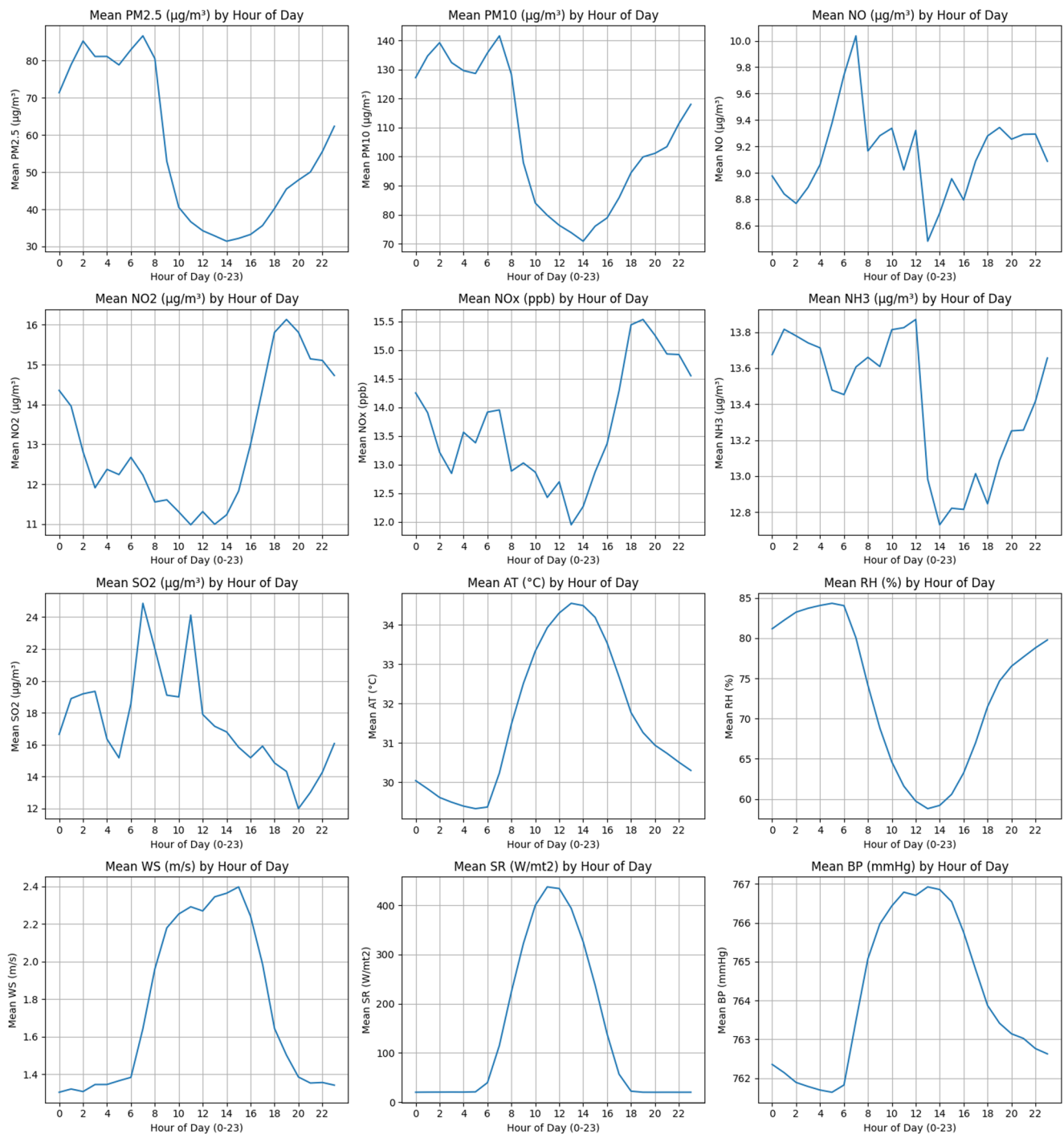


Fig. 3 Diurnal variation of pollutants and meteorological factors

The weather variables show opposite patterns: the temperature rises in the middle of the day (12–14 h), while the relative humidity falls at the same time and then rises again at night, and this shows the inverse. During the middle of the day (10–16 h), the wind speed and sun radiation are at their strongest, which helps pollutants spread. These data indicate a significant correlation between particulate matter,

gaseous pollutants, anthropogenic activities, and boundary layer dynamics, with pollution episodes peaking during stagnant morning and evening circumstances that restrict dispersion (Yadav et al., 2019).

The weekly changes (Fig. 4) in air quality and weather reveal rather constant trends with small changes from day to day. This is likely because of ongoing industrial and

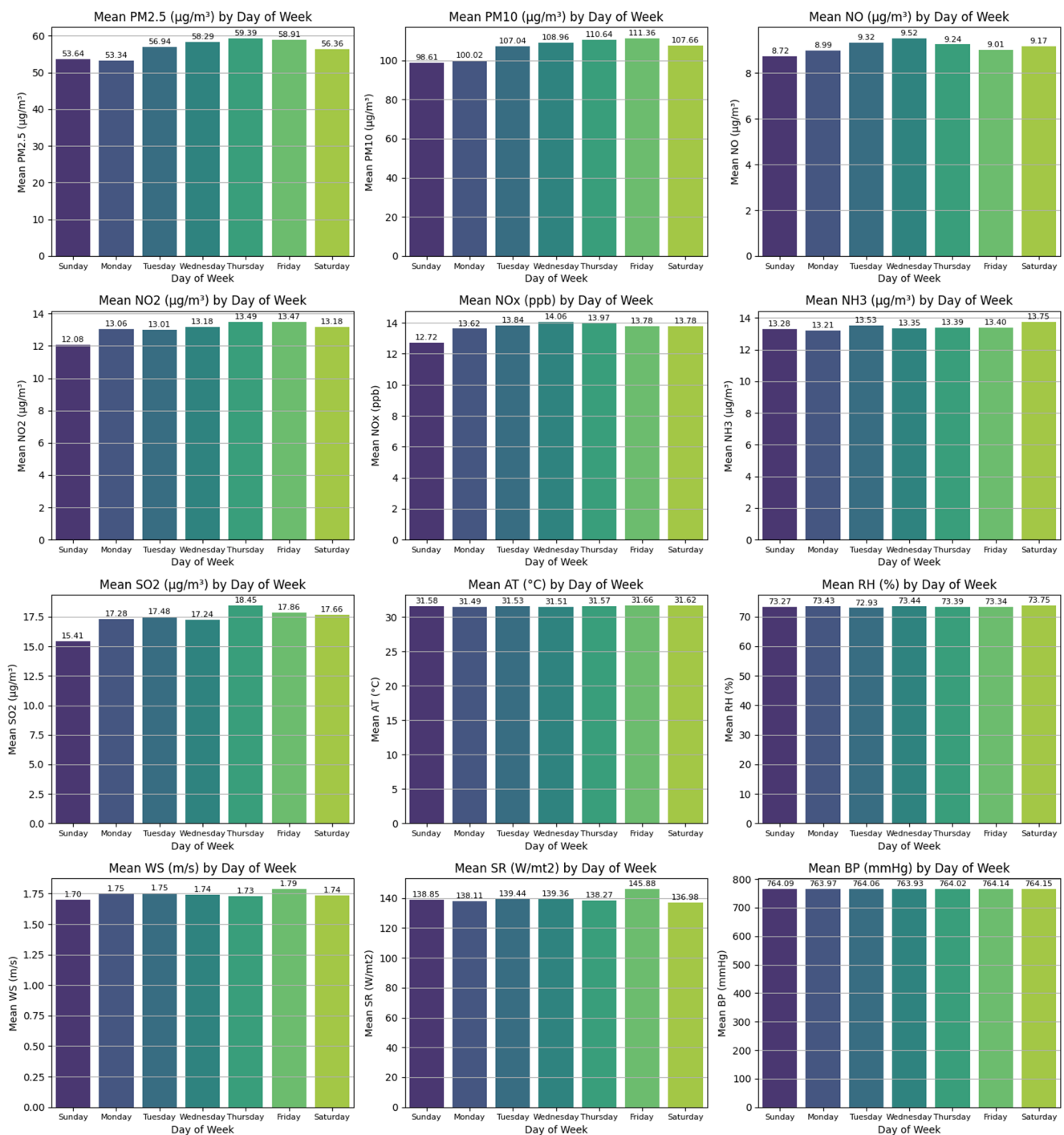


Fig. 4 Weekly variation of pollutants and meteorological factors

vehicle activity. PM_{2.5} levels stay high (about 53–56 $\mu\text{g}/\text{m}^3$), with the highest levels on Thursday. PM₁₀ levels are similar (around 96–113 $\mu\text{g}/\text{m}^3$), with the highest levels in the middle of the week (Wednesday–Friday). This suggests that more industrial activity and traffic on weekdays. Gaseous pollutants like NO, NO₂, and NO_x go up a little in the middle of the week compared to the weekend. This

suggests that humans are more active during the week. NH₃ and SO₂ show small changes, but they stay high the whole time. This shows that the sources of emissions are stable, perhaps because of industry and waste-related activities. The weather stays pretty much the same, with an average temperature of 31–32 $^{\circ}\text{C}$, a relative humidity of 72–74%, and a wind speed of about 1.7 m/s, which doesn't change

much from week to week. There are only small changes in solar energy and atmospheric pressure from day to day. In general, the data show that air pollution in Gummidipoondi is more affected by ongoing industrial activity and traffic emissions than by natural weekly cycles. On weekdays, the number of pollutants is a little higher than on weekends.

The monthly variations (Fig. 5) in pollutants and weather show strong seasonal patterns that are affected by

the monsoon and industrial emissions. The highest levels of $PM_{2.5}$ and PM_{10} are in the winter (January to February and November to December), when the average levels are over $80\text{--}100\text{ }\mu\text{g}/\text{m}^3$ for $PM_{2.5}$ and $130\text{--}150\text{ }\mu\text{g}/\text{m}^3$ for PM_{10} . During the southwest monsoon period (June to September), when it rains more and the wind blows more, the levels drop rapidly because the rain and wind wash away and dilute the particles (Sasmita et al., 2022). This is when

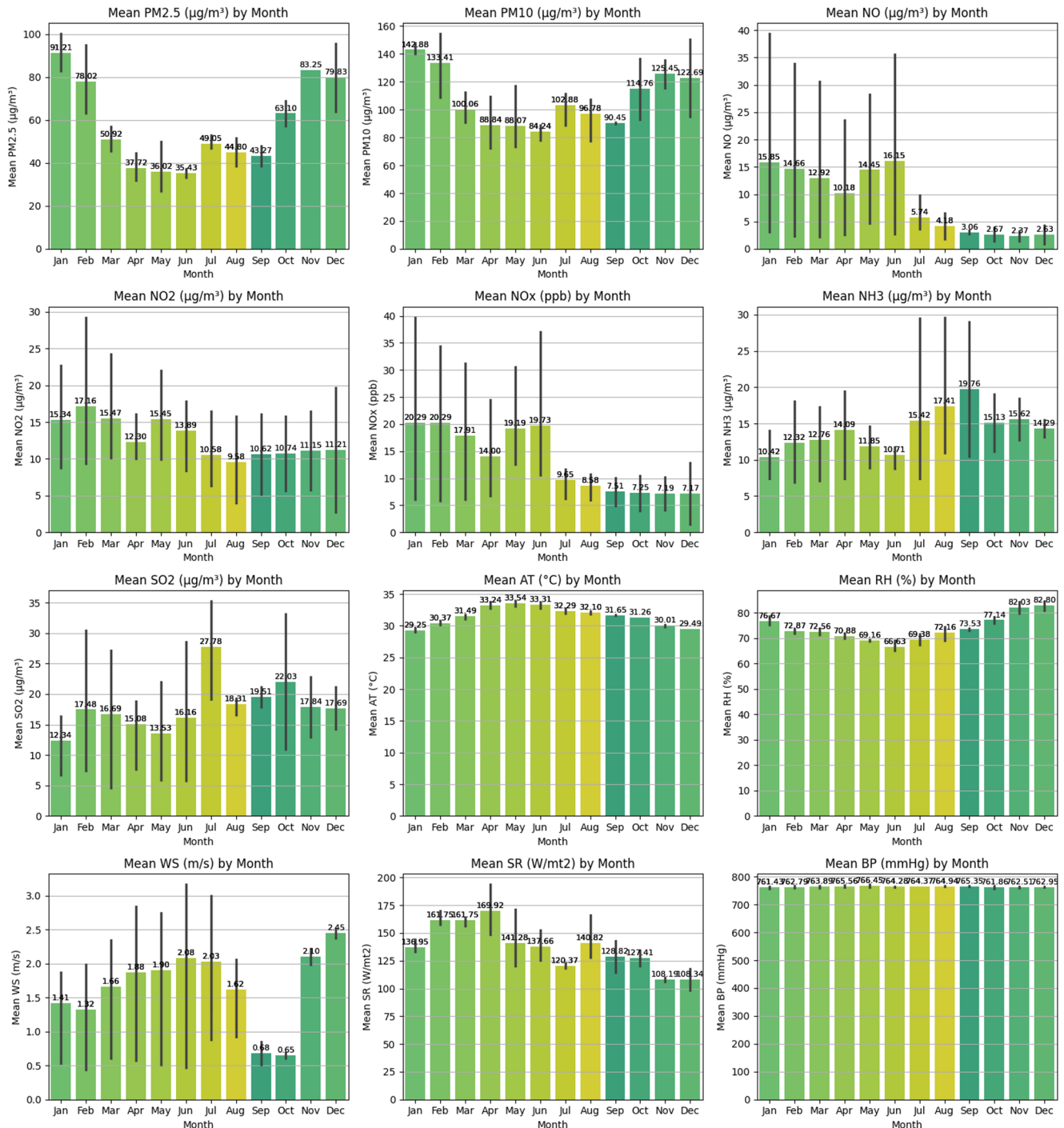


Fig. 5 Monthly variation of pollutants and meteorological factors

the concentrations are lowest (around 40 to 50 $\mu\text{g}/\text{m}^3$ for $\text{PM}_{2.5}$). NO , NO_2 , and NO_x all have similar seasonal patterns. Their levels are higher in the winter and early summer and lower during the monsoon months. This suggests that they are strongly linked to sources of combustion from cars and factories, as well as seasonal dispersion effects. SO_2 and NH_3 stay quite consistent, but they do have some peaks from May to July. These are probably caused by industrial activities and emissions from farms. Meteorological metrics exhibit anticipated seasonal trends: temperature reaches its peak in May–June (about 34 °C) and diminishes throughout the monsoon and winter (approximately 26–29 °C), while relative humidity peaks in October–December (exceeding 80%) due to northeast monsoon precipitation. Wind speed rises from June to August (up to 2.9 m/s), which helps pollutants spread out. In winter, on the other hand, wind speed drops (below 1 m/s), which causes pollutants to elevate. During the summer (March to June), solar radiation is at its highest, and during the monsoon, it is at its lowest. Barometric pressure is at its highest in December and January and at its lowest in the summer, which is consistent with the region's climate. In general, the results show that winter months are the most polluted since the air doesn't move, while monsoon months clear the air naturally. This makes AQI very seasonal in this industrial area. The observed winter and post-monsoon pollution peaks parallel findings from (Sasmita et al., 2022) and (Yadav et al., 2019), highlighting atmospheric stagnation and weak dispersion as universal factors in seasonal air quality deterioration.

The yearly comparison (Fig. 6) of air quality and weather data at Gummidipoondi shows both improvements and new problems from 2023 to 2025. $\text{PM}_{2.5}$ went down from 62.11 $\mu\text{g}/\text{m}^3$ in 2023 to 53.10 $\mu\text{g}/\text{m}^3$ in 2024, but then went back up to 55.19 $\mu\text{g}/\text{m}^3$ in 2025. This shows that there was some control, but not a long-term decrease. PM_{10} went down more quickly, from 120.97 $\mu\text{g}/\text{m}^3$ in 2023 to 95.19 $\mu\text{g}/\text{m}^3$ in 2024, and then to 104.89 $\mu\text{g}/\text{m}^3$ in 2025. This shows that particulate emissions were changing, probably because of industrial activity. In 2024, NO and NO_x levels declined a lot ($\text{NO}=3.37 \mu\text{g}/\text{m}^3$, $\text{NO}_x=9.52 \text{ ppb}$) compared to 2023. However, both levels went high in 2025, which could mean that emissions were controlled at times or that industrial and traffic levels were different. NH_3 , on the other hand, went down from 17.83 $\mu\text{g}/\text{m}^3$ in 2023 to 10.13 $\mu\text{g}/\text{m}^3$ in 2024, then increased to 12.79 $\mu\text{g}/\text{m}^3$ in 2025. SO_2 , on the other hand, went up consistently from 11.26 $\mu\text{g}/\text{m}^3$ in 2023 to 24.23 $\mu\text{g}/\text{m}^3$ in 2025. This suggests that sulphur emissions are getting worse, probably because of companies that burn fossil fuels and waste. The weather stayed rather consistent, with temperatures averaging 31–32 °C, relative humidity about 71–74%, and wind speeds going up in 2024 (2.21 m/s) compared to 2023 (0.93 m/s), then going down in

2025 (2.12 m/s). There was only a small amount of change in solar radiation and barometric pressure from year to year. Overall, the results show that 2024 had the best air quality because there were fewer particulate and gaseous pollutants. However, the fact that levels of PM_{10} , NO_x , NH_3 , and SO_2 went high in 2025 shows that we need to keep using strict and more consistent pollution control measures to keep air quality improvements in this industrial corridor.

The seasonal hourly changes (Fig. 7) show how both human activity and monsoon-driven weather affect air quality. In the winter, $\text{PM}_{2.5}$ and PM_{10} levels are highest, with peaks in the early morning (0–8 h) and late at night. This is because the boundary layer is low and the atmosphere is stable. The lowest levels of $\text{PM}_{2.5}$ and PM_{10} are during the southwest monsoon, when rain and stronger winds help spread and wash out the particles. NO , NO_2 , and NO_x all have similar seasonal trends. They are higher in the winter and after the monsoon because stable air conditions trap traffic and industrial emissions. They are lower during the monsoon because they are diluted. NH_3 stays quite steady throughout the year, however, it goes up a little during the monsoon, which may be because of farming. During the winter and after the monsoon, SO_2 levels change the most. This could be because there is more burning and less spreading. The weather patterns match the regional climate, temperatures are highest in the afternoons of the southwest monsoon and post-monsoon (up to around 35 °C), while winters are colder (about 28–30 °C). The southwest monsoon has the strongest winds (>2.5 m/s), which help pollutants disperse, while the winter has the lightest winds (<1.5 m/s), which help pollutants to trap. Solar radiation has a regular daily cycle that stays the same all year, with the most intense radiation at noon. However, during the monsoon months, clouds block some of the radiation. The highest barometric pressure is in the winter and after the monsoon, while the lowest is in the summer and during the southwest monsoon. The overall results show that winter is the season with the most pollution because the air doesn't move, while the southwest monsoon brings the cleanest air because it spreads out naturally and picks up trash, showing how strongly meteorology affects pollution levels in this industrial cluster.

Figure 8 illustrates the monthly variation of average AQI. January has the highest average AQI of 196.84, followed by November (174.15) and December (159.54). This shows that pollution levels are very high in the winter because of stable weather, low wind speeds, and more industrial and combustion activity. June has the cleanest air (AQI=81.45), and the AQI values stay low from May to September. This is because the monsoon cleans the air by bringing rain and spreading it out more. This cycle shows that AQI is highest in the winter and after the rainfall, but the monsoon months give some relief. Overall, the results show that the AQI in

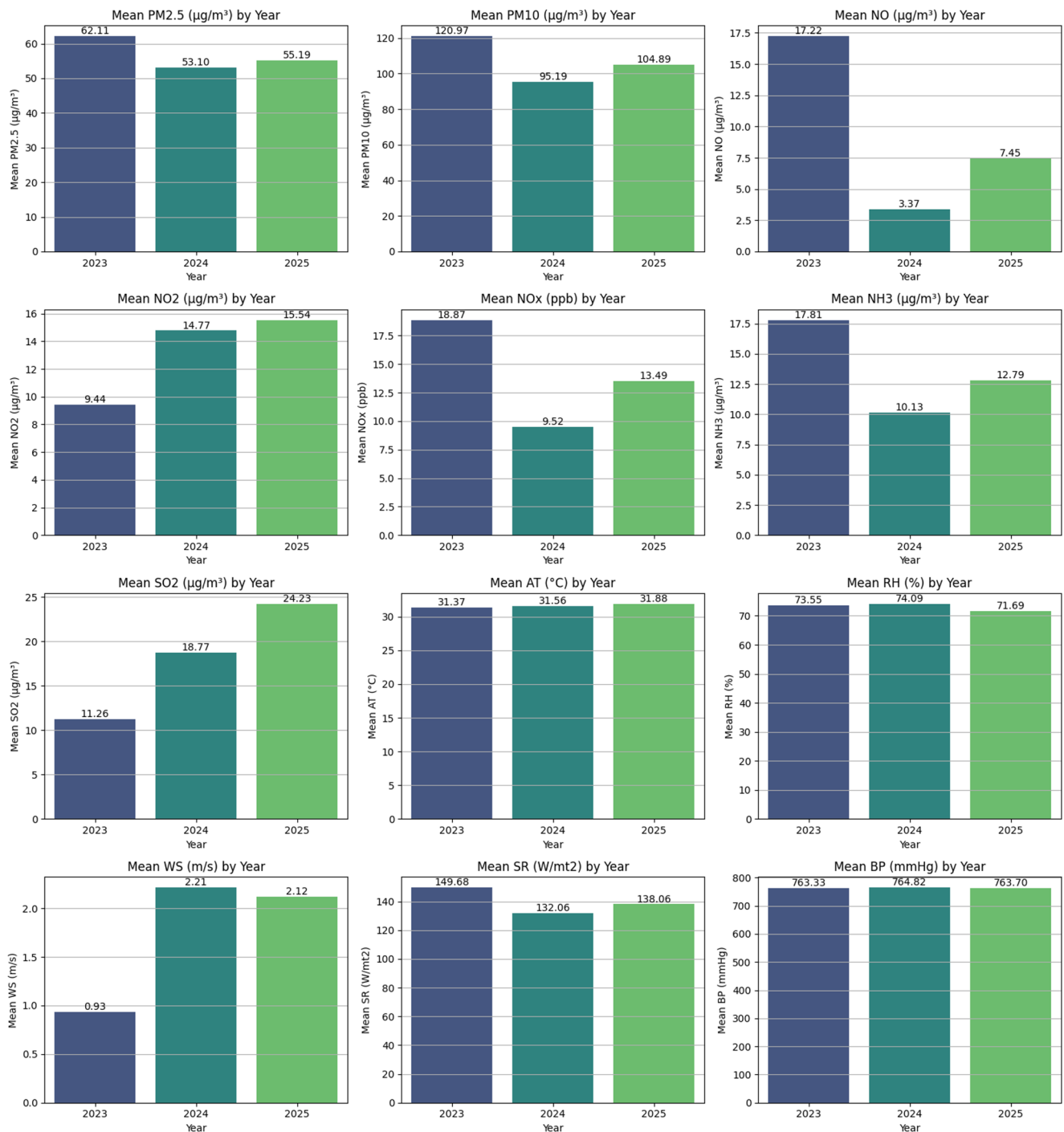


Fig. 6 Annual comparison of air quality and meteorological parameters (2023–2025)

this industrial cluster is greatly affected by both the weather and human activities, with winter and weekdays being the most dangerous times for air quality.

Machine learning models for AQI predictions

The examination of various machine learning models for AQI prediction and model performance metrics (Table 4)

indicates that the XGBRegressor had the best predictive accuracy ($R^2 = 0.9965$) and the lowest error values (MAE=0.0223, RMSE=0.0573). Its actual vs. predicted plot (Fig. 9) was very close to the 45° line, and the residuals (Fig. 10) were very close to zero, showing that it was both accurate and reliable. The superior accuracy of XGBoost in this study aligns with results from (Su et al., 2023) in Beijing and recent Indian studies (de Bont et al., 2024), confirming

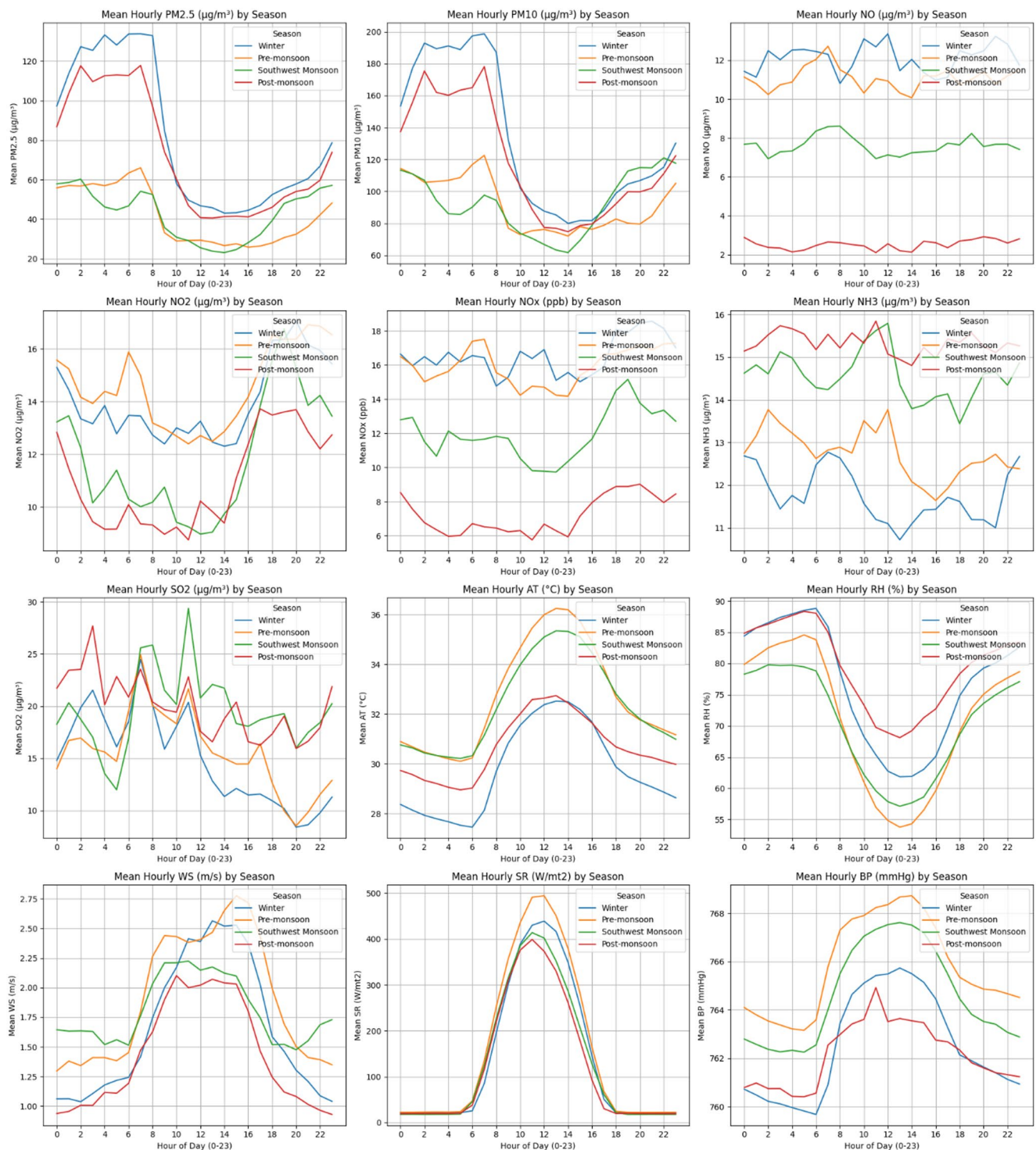


Fig. 7 Seasonal hourly variation of pollutants and meteorological factors

the robustness of tree-based ensemble approaches for non-linear, multi-variable AQI prediction. Although XGBoost achieved a very high R^2 (0.9965), steps were taken to check for overfitting. A k -fold cross-validation strategy ($k=X$) was employed, ensuring that model performance was consistent across folds. Residual analysis and REC curves further

confirmed that prediction errors remained uniformly low across the dataset, indicating strong generalisation capability rather than overfitting. The RandomForestRegressor predicted very well ($R^2 = 0.9946$, $\text{RMSE} = 0.0712$), although its residuals were a little more spread out than XGB's. However, it still had strong generalisation. Ridge regression and

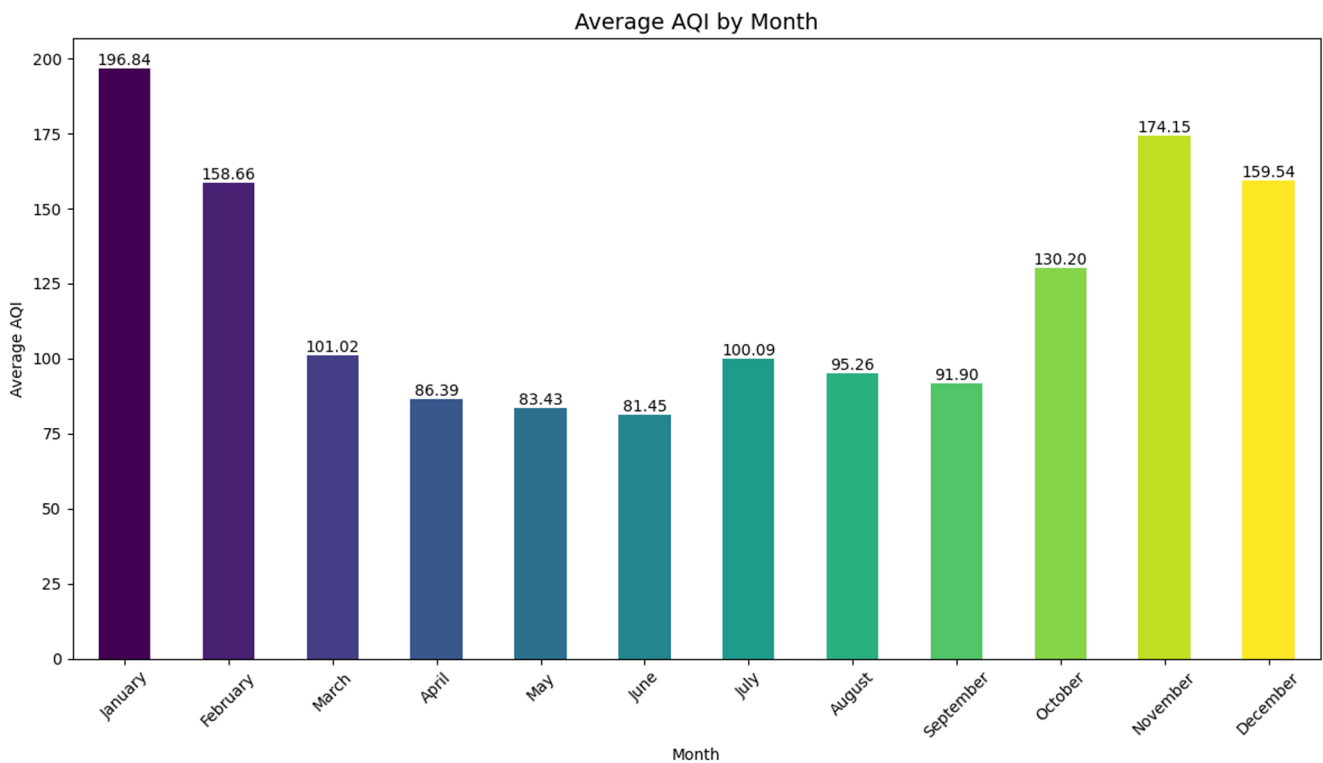


Fig. 8 Average AQI Values (monthly)

Table 4 Model evaluation metrics (R^2 , MAE, MSE, RMSE) for different machine learning models on AQI prediction

Machine Learning Models	R -squared	MAE	MSE	RMSE
Ridge	0.9575	0.1574	0.0398	0.1994
RandomForestRegressor	0.9946	0.0229	0.0051	0.0712
XGBRegressor	0.9965	0.0223	0.0033	0.0573
SVR	0.9579	0.1525	0.0394	0.1985
KNeighborsRegressor	0.7614	0.3422	0.2234	0.4726
MLPRegressor	0.9797	0.1091	0.0189	0.1376

SVR, on the other hand, showed a good fit to the data ($R^2 = 0.958$), but their residuals were more spread out, especially at extreme levels. This suggests that they may not be able to capture the non-linear relationships that are a part of how air quality changes. The KNeighborsRegressor had the worst performance ($R^2 = 0.7614$, $RMSE = 0.4726$), with large differences in predictions and residuals. This means it is not good for predicting AQI in high dimensions. The MLPRegressor had a fair level of accuracy, but the spread of the residuals showed that it wasn't always able to generalise well. The investigation reveals that tree-based ensemble models, especially XGBoost, are better for AQI modelling because they can handle complicated, non-linear interactions between pollutants and weather.

The XGBoost model's feature importance analysis (Fig. 11) makes it evident that particulate matter is the main feature in the AQI prediction. $PM_{2.5}$ is the most important (0.7042), followed by PM_{10} (0.2933). These two

characteristics make up almost all of the model's predictive efficiency, which shows how important they are for figuring out air quality levels. This study's results validate the central role of $PM_{2.5}$ and PM_{10} in AQI variation, corroborating the conclusions of (Joshi et al., 2025), (Pant et al., 2024), and multicity work by (de Bont et al., 2024), where particulate matter consistently emerged as the strongest predictor and health risk factor for Indian and South Asian urban environments. Other pollutants and weather conditions, like SO_2 , NH_3 , NO_2 , and RH, have very little effect on AQI prediction (all < 0.001). Variables like temperature, wind speed, solar radiation, barometric pressure, and NO_x have essentially no effect on AQI prediction. This shows how much AQI depends on the concentrations of fine and coarse particles. It also shows how important it is to adopt targeted mitigation measures that focus on reducing particulate matter to make a big difference in air quality.

The Regression Error Characteristic (REC) curve (Fig. 12) comparison clearly shows that XGBoost, followed closely by RandomForestRegressor and Ridge regression, has the best predictive accuracy for AQI in the Gummidipoondi dataset. This is shown by their curves reaching the upper-left region of the plot with very little error. This means that a much larger percentage of their forecasts fall within tiny absolute error margins than those of other models. The KNeighborsRegressor, on the other hand, has the worst performance. It needs far greater error tolerances to

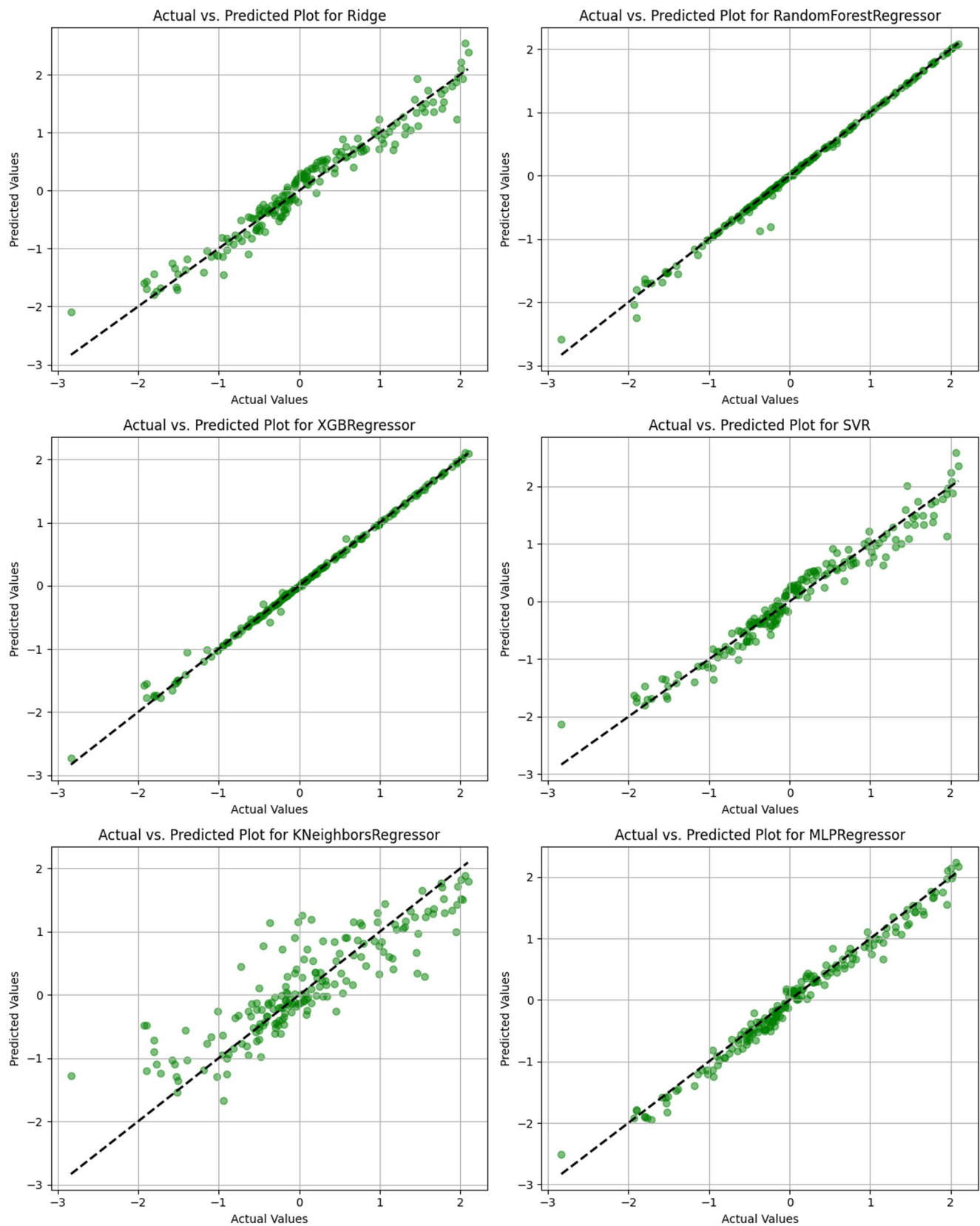


Fig. 9 Actual vs. predicted AQI plots for machine learning models

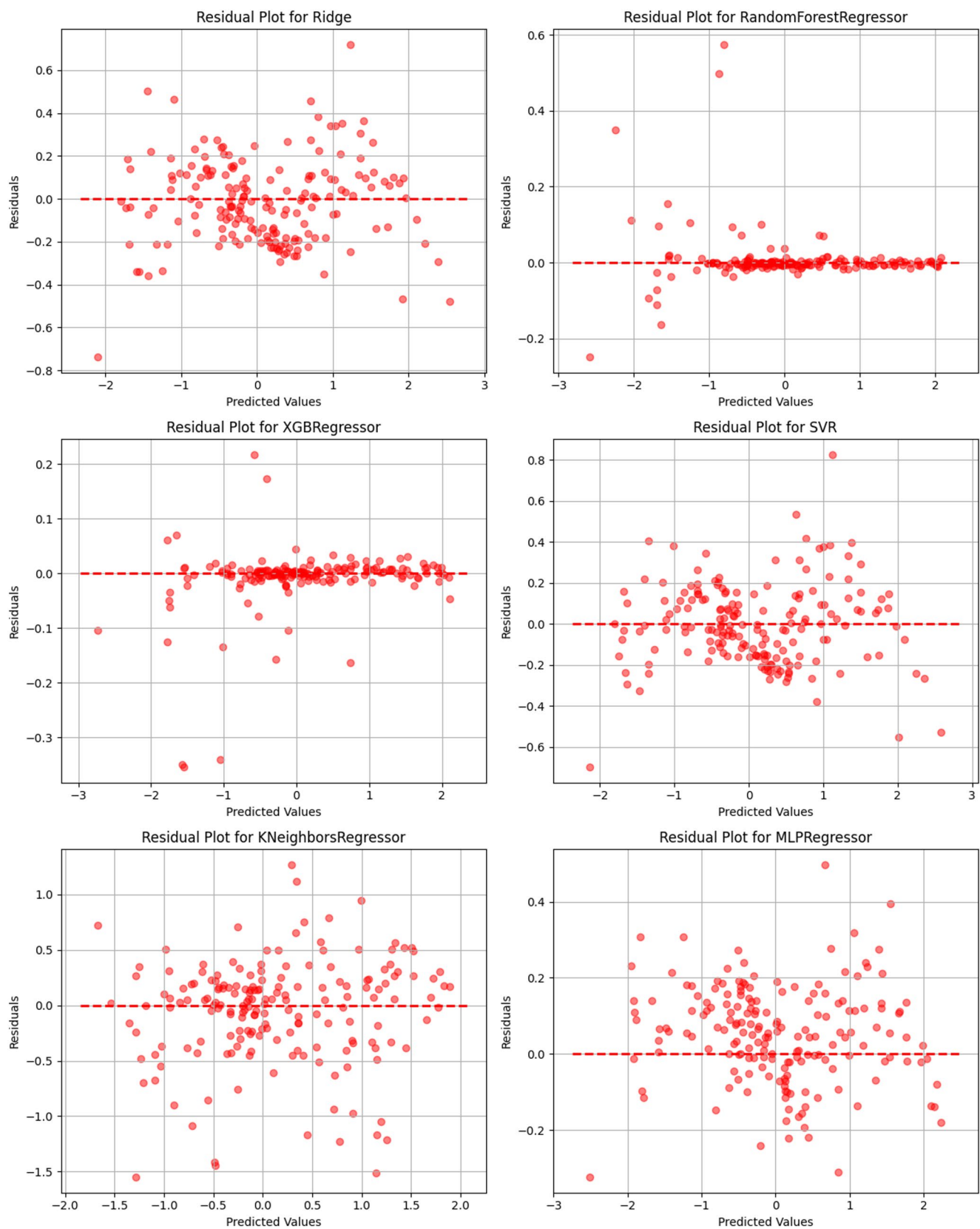


Fig. 10 Residual plots of machine learning models for AQI prediction

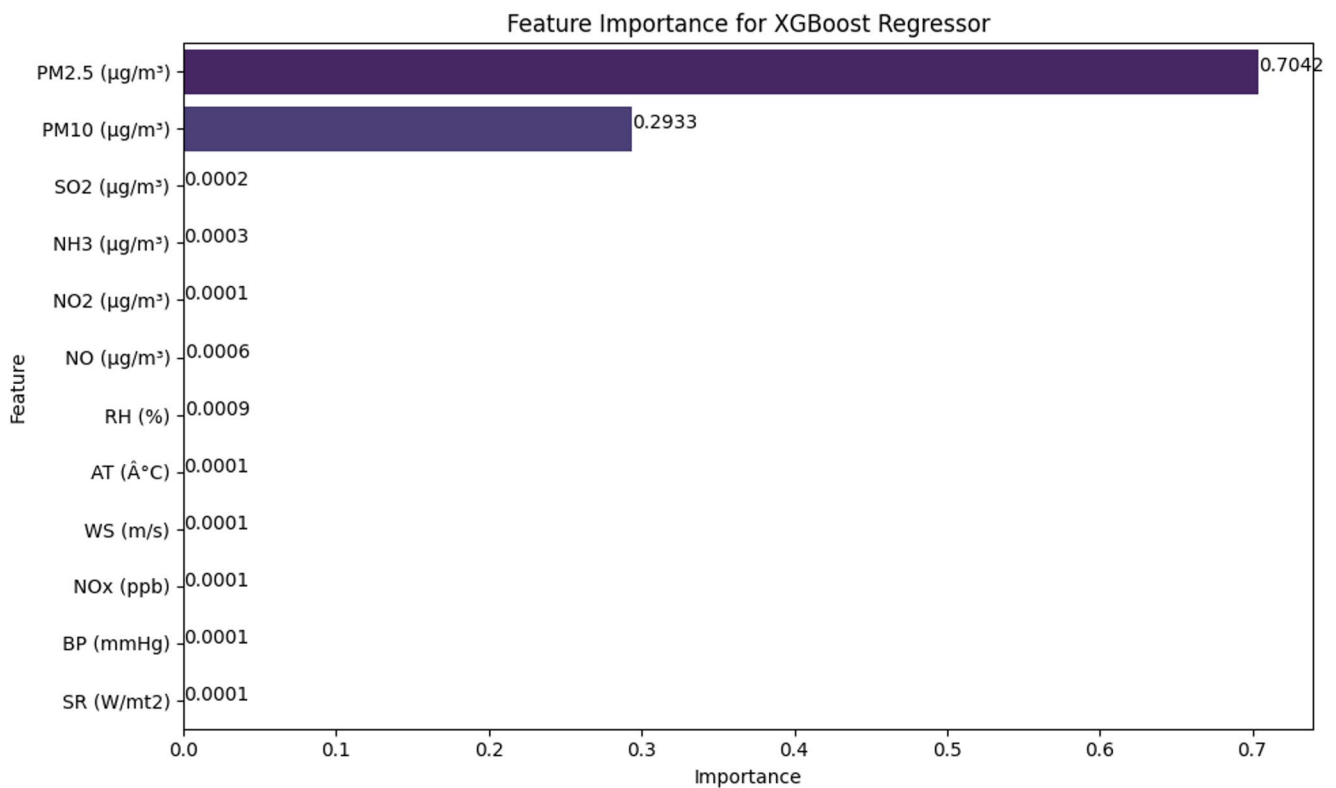


Fig. 11 Feature importance analysis from XGBoost model

get the same cumulative proportion of accurate predictions. The MLPRegressor and SVR come next, with intermediate but not great accuracy. The REC analysis strongly supports the idea that ensemble-based and regularised linear models are the best choice for this industrial air quality context. This is because they are good at capturing complicated pollution patterns and are therefore a good choice for real-time AQI forecasting and policymaking in peri-urban industrial areas.

The SHAP summary plot (Fig. 13) shows that $PM_{2.5}$ and PM_{10} are the most important factors that affect AQI predictions. $PM_{2.5}$ has the strongest positive and negative effects on model output. When $PM_{2.5}$ levels are high (red), AQI projections always go up, which means that air quality is getting worse. When $PM_{2.5}$ levels are low (blue), AQI predictions go down. PM_{10} also plays a significant role, but not as much as $PM_{2.5}$. In contrast, other pollutants (SO_2 , NO_2 , NO , NH_3) and climatic factors (temperature, humidity, wind speed, solar radiation, pressure) have SHAP values that are low and clustered around zero, indicating a little predictive influence on AQI within this dataset. This confirms that particulate matter, especially fine particles, is the most important factor in determining air quality, which supports both the feature importance analysis and the need for PM reduction in real life to improve air quality.

The autocorrelation plot (Fig. 14) of AQI during a 30-day lag period shows that the autocorrelation stays high and

slowly goes down. It starts at 1.0 for lag 0 and stays over 0.3 even at lag 30 days. This means that AQI values in Gummidipoondi depend a lot on the time, with air quality levels on any particular day being highly affected by those from the days before. Such long-range persistence is due to long-term emissions from industry and traffic, as well as slow atmospheric dispersion processes during periods of stagnation, especially in the winter and after the monsoon. The slow decline indicates the accumulation of contaminants and the existence of underlying periodicities or cycles in AQI, highlighting the need for ongoing monitoring and the significance of time.

The LIME local explanation plot (Fig. 15) shows how each feature affected the AQI prediction for a certain case in the Gummidipoondi dataset. Notably, high $PM_{2.5}$ levels were the biggest factor in raising the expected AQI, which shows that tiny particulate matter is the main cause of poor air quality in the area. In this case, greater PM_{10} levels had the opposite effect on the AQI prediction, probably because the model was able to take into account the specific context of the data point. Relative humidity and SO_2 , together with NO , also made the prediction increase a little, while wind speed, barometric pressure, and NO_x made the projected AQI decrease. Figure 15 shows that fine particulate concentrations are still the most important factor for local air quality risk, even when other meteorological and secondary

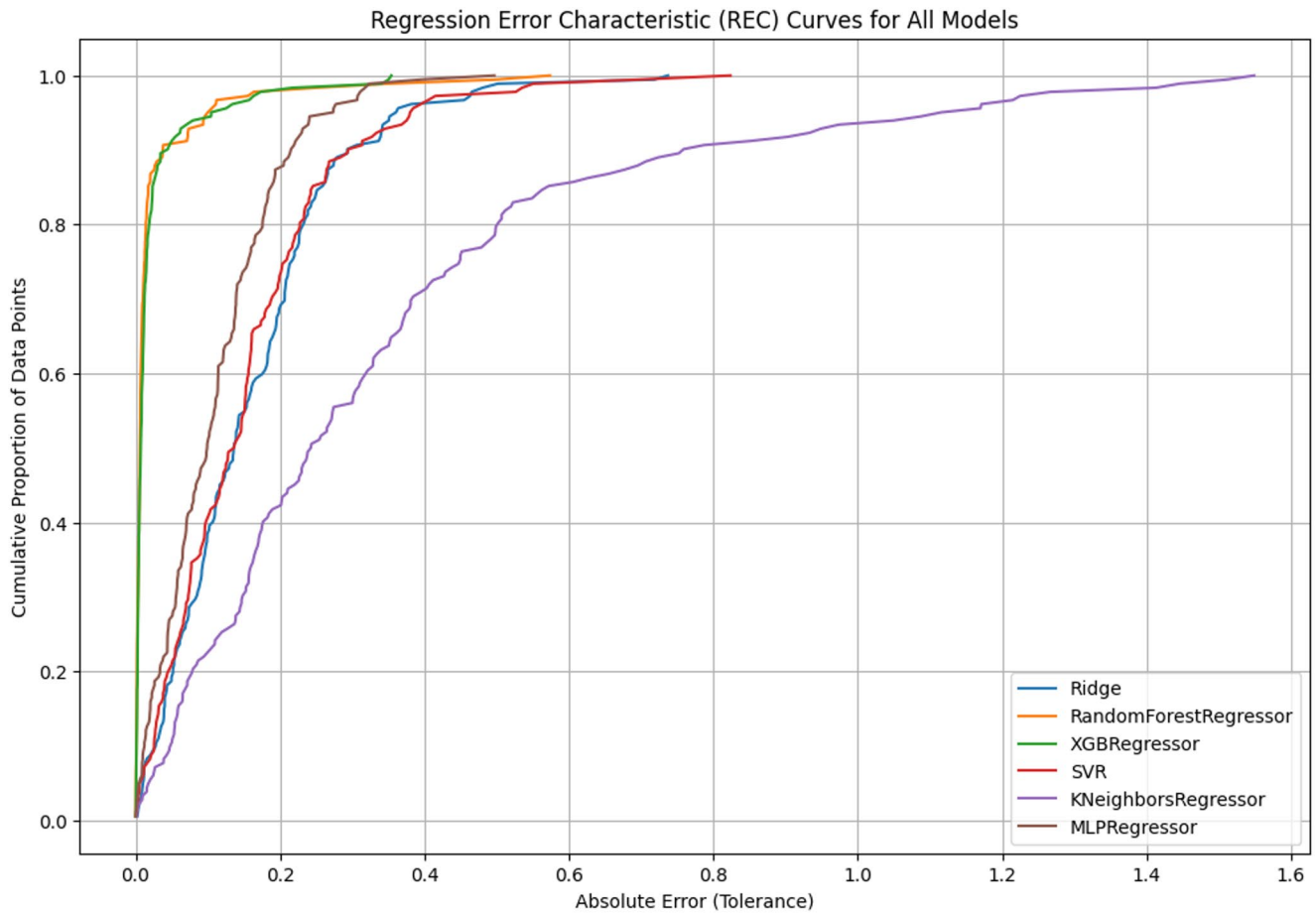


Fig. 12 Regression Error Characteristic (REC) curves for all machine learning models

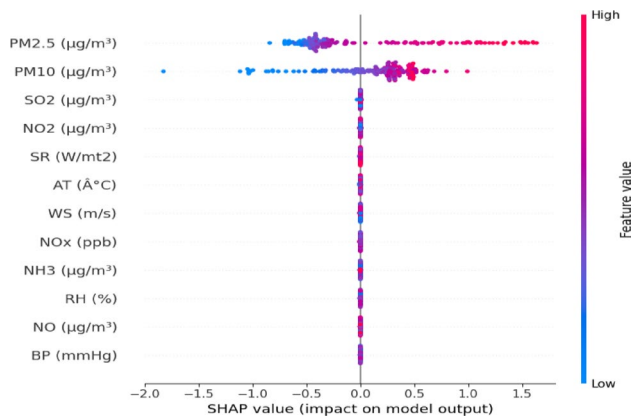


Fig. 13 SHAP summary plot AQI prediction

pollutants are present. This provides a clear and understandable reason for the model outputs, which is important for targeted interventions and clear policy communication.

Conclusion

This study shows that fine particulate matter is the main cause of poor air quality in Gummidipoondi. Controlling $PM_{2.5}$ is the best strategy to improve the AQI. Temporal assessments highlighted the concerns present during winter and post-monsoon months when air stagnation exacerbates pollutant accumulation. The use of advanced ensemble models showed that machine learning can make very accurate and clear predictions that can be used in decision-support systems. Policy action should put the strict regulation of industrial emissions, the good management of vehicle traffic along freight routes, and the building of more infrastructure for continuous monitoring at the top of the list. Integrating AI-driven prediction frameworks into regional governance can facilitate early warning systems, mitigate exposure risks, and inform long-term strategic planning. Future research should encompass multi-site industrial clusters, incorporate satellite-based monitoring, and assess health burden correlations to enhance the evidentiary base for sustainable air quality management in South India. While our findings fit well with large-city research, a limitation is

Fig. 14 Autocorrelation plot of AQI over a 30-day lag

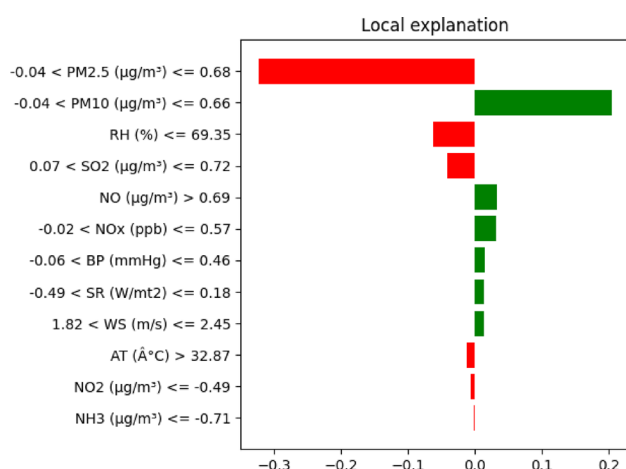
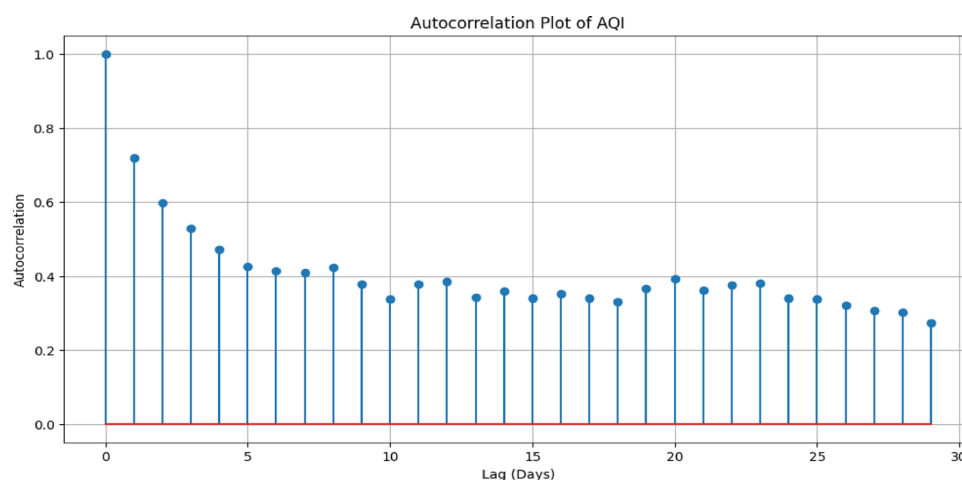


Fig. 15 LIME-based local explanation of AQI prediction

the relatively short monitoring duration and single-site context, which might not capture spatial heterogeneity seen in multi-cluster studies. These limitations suggest directions for future work, including expanding to multi-site, multi-year datasets and integrating satellite-based observations, as recommended by recent reviews.

Author contributions R. Muralikrishnan: Conceptualization, Methodology, Supervision, Writing – Original Draft, Project Administration. Suneel Gollapalli: Data Curation, Software Implementation, Formal Analysis, Visualization, Writing – Review & Editing. Elayaraja Sellappan: Methodology, Model Development, Supervision, Writing – Review & Editing. Prasanya J: Data Collection, Preprocessing, Investigation, Validation. V. Murali: Literature Review, Statistical Analysis, Resources, Data Interpretation. A. Vijayakumar: Model Evaluation, Technical Review, Writing – Final Review & Proofreading.

Funding No funding was received.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

References

- Andújar-Maqueda, J., Ortiz-Amezcu, P., Cariñanos, P., Abril-Gago, J., De Linares, C., de Arruda Moreira, G., Bravo-Aranda, J. A., Granados-Muñoz, M. J., Alados-Arboledas, L., & Guerrero-Rascado, J. L. (2025). The role of atmospheric boundary layer wind and turbulence on surface pollen levels. *Agricultural and Forest Meteorology*, 371, 110584. <https://doi.org/10.1016/J.AGRFORMET.2025.110584>
- de Bont, J., Krishna, B., Stafoggia, M., Banerjee, T., Dholakia, H., Garg, A., Ingole, V., Jaganathan, S., Kloog, I., Lane, K., Mall, R. K., Mandal, S., Nori-Sarma, A., Prabhakaran, D., Rajiva, A., Tiwari, A. S., Wei, Y., Wellenius, G. A., Schwartz, J., & Lungman, P. (2024). Ambient air pollution and daily mortality in ten cities of India: A causal modelling study. *The Lancet Planetary Health*, 8(7), e433–e440. [https://doi.org/10.1016/S2542-5196\(24\)00114-1](https://doi.org/10.1016/S2542-5196(24)00114-1)
- Joshi, D. C., Negi, P., Devi, S., Lohani, H., Kumar, R., Gupta, M., & Ming, L. C. (2025). Fine particulate matter (PM2.5, PM10): A silent catalyst for chronic lung diseases in India; a comprehensive review. *Environmental Challenges*, 20, 101215. <https://doi.org/10.1016/J.ENVC.2025.101215>
- K, K., S.K, A., R, D., & Ravindiran, G. (2024). Integrating machine learning techniques for air quality index forecasting and insights from pollutant-meteorological dynamics in sustainable urban environments. *Earth Science Informatics*, 17(4), 3733–3748. <https://doi.org/10.1007/S12145-024-01382-8/METRCS>
- Kalbarczyk, R., & Kalbarczyk, E. (2020). Meteorological conditions of the winter-time distribution of nitrogen oxides in Poznań: A proposal for a catalog of the pollutants variation. *Urban Climate*, 33, 100649. <https://doi.org/10.1016/J.UCLIM.2020.100649>
- Kaveh, A. (2024). Applications of artificial neural networks and machine learning in civil engineering. *Studies in Computational Intelligence*, 1168, 1–474. <https://doi.org/10.1007/978-3-031-66051-1/COVER>
- Kaveh, A., & Khavaninzadeh, N. (2024). Shear strength of cellular steel beams predicted by hybrid ANFIS-ECBO model. *Arabian Journal for Science and Engineering*, 1–22. <https://doi.org/10.1007/S13369-024-09802-Z/TABLES/5>
- Kaveh, A., Bakhshpoori, T., & Hamze-Ziabari, S. M. (2018). M5^{*} and Mars based prediction models for properties of Self-Compacting

- concrete containing fly Ash. *Periodica Polytechnica Civil Engineering*, 62(2), 281–294. <https://doi.org/10.3311/PPCI.10799>
- Liu, B. H., Zhang, L. W., Wei, Y. Q., & Chen, C. (2024). Dual power transformation and Yeo–Johnson techniques for static and dynamic reliability assessments. *Buildings* 2024, 14(11), 3625. <https://doi.org/10.3390/BUILDINGS14113625>. 14.
- Manisalidis, I., Stavropoulou, E., Stavropoulos, A., & Bezirtzoglou, E. (2020). Environmental and health impacts of air pollution: A review. *Frontiers in Public Health*, 8, 505570. <https://doi.org/10.3389/FPUH.2020.00014/BIBTEX>
- Meena, K. K., Bairwa, D., & Agarwal, A. (2024). A machine learning approach for unraveling the influence of air quality awareness on travel behavior. *Decision Analytics Journal*, 11, 100459. <https://doi.org/10.1016/J.DAJOUR.2024.100459>
- Mujtaba, M. A., Munir, M. A., Ali, S., Petrù, J., Ansar, T., Akhlaq, W., Ahmad, M., Iqbal, H., Ali, F., Bashir, M. N., & Alexander, T. (2025). Using machine learning for air quality prediction and sustainable urban planning. *Sustainable Futures*, 10, 100981. <https://doi.org/10.1016/J.SFTR.2025.100981>
- Pant, A., Sharma, S., & Pant, K. (2023). Evaluation of machine learning algorithms for air quality index (AQI) prediction. *Journal of Reliability and Statistical Studies*, 16(2), 229–242. <https://doi.org/10.13052/JRSS0974-8024.1621>
- Pant, A., Joshi, R. C., Sharma, S., & Pant, K. (2024a). Air quality and public health risk assessment: A case of an industrial area in Haridwar City, Uttarakhand (India). *Indian Journal of Public Health*, 68(2), 222–226. https://doi.org/10.4103/IJPH.IJPH_279_23
- Pant, A., Pant, K., Pathak, N., & Ram, M. (2024b). Prediction of particulate matter (PM_{2.5}) for industrial area based on Naive Bayes classifier. *Lecture Notes in Networks and Systems*, 786, 189–195. https://doi.org/10.1007/978-981-99-6547-2_15
- Payus, C. M., Syazni, N., M. S., & Sentian, J. (2022). Extended air pollution index (API) as tool of sustainable indicator in the air quality assessment: El-Nino events with climate change driven. *Heliyon*, 8(3), e09157. <https://doi.org/10.1016/J.HELİYON.2022.E09157>
- Raiaan, M. A. K., Sakib, S., Fahad, N. M., Mamun, A., Al, Rahman, M. A., Shatabda, S., & Mukta, M. S. H. (2024). A systematic review of hyperparameter optimization techniques in convolutional neural networks. *Decision Analytics Journal*, 11, 100470. <https://doi.org/10.1016/J.DAJOUR.2024.100470>
- Ravindiran, G., Rajamanickam, S., Kanagarathinam, K., Hayder, G., Janardhan, G., Arunkumar, P., Arunachalam, S., AlObaid, A. A., Warad, I., & Muniasamy, S. K. (2023). Impact of air pollutants on climate change and prediction of air quality index using machine learning models. *Environmental Research*, 239, 117354. <https://doi.org/10.1016/J.ENVRES.2023.117354>
- Ravindiran, G., Karthick, K., Rajamanickam, S., Datta, D., Das, B., Shyamala, G., Hayder, G., & Maria, A. (2025). Ensemble stacking of machine learning models for air quality prediction for Hyderabad City in India. *IScience*, 28(2), 111894. <https://doi.org/10.1016/j.isci.2025.111894>
- Sananmuang, T., Mankong, K., & Chokeshaiusaha, K. (2024). Multi-layer perceptron and support vector regression models for feline parturition date prediction. *Heliyon*, 10(6), e27992. <https://doi.org/10.1016/J.HELİYON.2024.E27992>
- Sasmitha, S., Kumar, D. B., & Priyadharshini, B. (2022). Assessment of sources and health impacts of PM₁₀ in an urban environment over Eastern coastal plain of India. *Environmental Challenges*, 7, 100457. <https://doi.org/10.1016/J.ENV.2022.100457>
- Singh, V., Singh, S., Biswal, A., Kesarkar, A. P., Mor, S., & Ravindra, K. (2020). Diurnal and Temporal changes in air pollution during COVID-19 strict lockdown over different regions of India. *Environmental Pollution*, 266, 115368. <https://doi.org/10.1016/J.ENV.2020.115368>
- Su, M., Liu, H., Yu, C., & Duan, Z. (2023). A novel AQI forecasting method based on fusing Temporal correlation forecasting with Spatial correlation forecasting. *Atmospheric Pollution Research*, 14(4), 101717. <https://doi.org/10.1016/J.APR.2023.101717>
- Tao, H., Jawad, A. H., Shather, A. H., Al-Khafaji, Z., Rashid, T. A., Ali, M., Al-Ansari, N., Marhoon, H. A., Shahid, S., & Yaseen, Z. M. (2023). Machine learning algorithms for high-resolution prediction of Spatiotemporal distribution of air pollution from meteorological and soil parameters. *Environment International*, 175, 107931. <https://doi.org/10.1016/J.ENVINT.2023.107931>
- Yadav, R., Sahu, L. K., Beig, G., Tripathi, N., Maji, S., & Jaaffrey, S. N. A. (2019). The role of local meteorology on ambient particulate and gaseous species at an urban site of Western India. *Urban Climate*, 28, 100449. <https://doi.org/10.1016/J.UCLIM.2019.01.003>
- Zalakeviciute, R., Rybarczyk, Y., Granda-Albuja, M. G., Diaz Suarez, M. V., & Alexandrino, K. (2020). Chemical characterization of urban PM₁₀ in the tropical Andes. *Atmospheric Pollution Research*, 11(2), 343–356. <https://doi.org/10.1016/J.APR.2019.1.007>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.